

Mark A. Kheyfets

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EDUCATION

Worcester Polytechnic Institute | Worcester, MA 01609

Major: Robotics Engineering

GPA: 3.6

Expected Graduation: *May 2027*

PROFESSIONAL EXPERIENCE

Research Analyst | *OpenStar Technologies*

Spring 2026

Conducted mixed-methods research on public sentiment toward nuclear fusion in New Zealand's energy sector.

Contributed to data analysis and visualization using Streamlit, identified key trends in fusion familiarity and public support, and helped develop an evidence-based outreach policy for OpenStar Technologies focused on public engagement, audience-first storytelling, and legislative outreach.

Resident Advisor | *Worcester Polytechnic Institute*

Fall 2025-Spring 2026

Selected to serve as a peer leader and resource for students within residential housing, fostering an inclusive and supportive community. Responsibilities include community-building, enforcing campus policies, crisis management, and facilitating residential programming to enhance student life and well-being.

PROJECTS

Vision Guided Robotic Pick-and-Place System:

Real-Time Object Detection and Autonomous Sorting using Computer Vision and Kinematics.

- Built an autonomous robotic sorting system using computer vision, forward and inverse kinematics, and quintic trajectory planning.
- Implemented masking, color thresholding, centroid detection, fisheye correction, and image-to-robot coordinate transformations for reliable object localization.
- Troubleshoot calibration, sensing, and motion-planning issues to improve execution accuracy and repeatability.

Romi Autonomous Navigation and AprilTag Detection System:

Differential-Drive Mobile Robot with Forward/Inverse Kinematics and Computer Vision.

- Implemented second-order forward kinematic pose estimation, trajectory control, and autonomous path planning for accurate real-time localization and navigation.
- Integrated OpenMV computer vision and AprilTag detection for autonomous target searching, alignment, and docking using proportional control and state-machine-based behaviors.
- Developed proportional control routines for tag alignment and autonomous docking, including searching and recovery behaviors when targets were lost.

Translational Parallel Manipulator:

Autonomous Warehouse Robot with Parallel Manipulator.

- Designed and programmed a 2-DOF translational parallel manipulator for autonomous box pick-and-place on multi-level warehouse shelves, integrating one brushed DC motor and two servo motors with inverse kinematics to maintain constant gripper orientation.
- Integrated one brushed DC motor and two servo motors using inverse kinematics to maintain constant parallel gripper orientation.
- Developed custom C++ firmware for Romi base navigation, manipulator motion, and gripper actuation via serial-based control.

SKILLS

- Python, Racket, MATLAB, C, C++
- MS Office, GitHub, Fusion 360, SolidWorks, Streamlit
- Inverse kinematics, Trajectory Planning, Image Processing, FEM Analysis, Sensor Integration
- Sensor Input Devices: Cameras, IR/Ultrasonic